

# How to Study This Course

## *A Step-by-Step Study Guide — Clinical Data Analysis in R (Phase I)*

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Not sure where to begin? You are in the right place. This guide takes you, one small step at a time, from a blank laptop to being ready for the first session. You do **not** need any programming experience — just follow the steps in order. Most take only a few minutes.

## 1. How the course is organised

The course lives in a few clearly named folders. This week you only need the first one.

| Folder              | What it is / when to use it                                                                               |
|---------------------|-----------------------------------------------------------------------------------------------------------|
| Course_Preparation/ | START HERE. Your self-study pack for the week before the course. Everything in this guide happens here.   |
| Course/             | The main course used during the live sessions (lecture materials, scripts, data). You do not need it yet. |
| Deck_and_Build/     | The slide decks and the tools that build them (for the trainers). You can ignore this.                    |

Inside **Course\_Preparation/** the folders are numbered in the exact order you should follow them:

| Folder                           | What you do there                             |
|----------------------------------|-----------------------------------------------|
| 00_READ_ME_FIRST/                | Read the welcome and this Study Guide.        |
| 01_Install_R_and_RStudio/        | Install the software (Steps 2–4 below).       |
| 02_Getting_Started_with_RStudio/ | A short tour of the RStudio screen.           |
| 03_R_Basics/ → 07_Regression/    | The five lessons, in order.                   |
| Data/                            | The practice dataset and its data dictionary. |
| Exercises/ and Solutions/        | Short practice tasks and their answers.       |
| Cheat_Sheets/ and Guides/        | Quick references to keep open while you work. |

## 2. Install R

R is the free engine that does the analysis. Install it first.

1. Go to <https://cran.r-project.org>
2. **Windows:** click *Download R for Windows* → *base* → *Download R-4.x.x for Windows*, then run the installer and accept the defaults.
3. **macOS:** click *Download R for macOS* and choose the package for your Mac (the **-arm64** one for Apple M1/M2/M3 chips; the Intel one for older Macs). Open the .pkg and follow the steps.

## 3. Install RStudio Desktop

RStudio is the friendly window you actually work in. Install R first, then RStudio.

1. Go to <https://posit.co/download/rstudio-desktop/>
2. Download the free **RStudio Desktop** for your system (Windows .exe or macOS .dmg) and install it.
3. Open **RStudio** (not plain R). RStudio runs R for you.

## 4. Install the R packages

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Packages are free add-ons that give R extra tools. Open RStudio, click in the **Console** (the bottom-left pane), paste the line below, and press Enter. It may take a few minutes.

```
install.packages(c("tidyverse", "readxl", "gtsummary",  
                  "broom", "survival", "survminer"))
```

**TIP** You only install a package once per computer. In every new session you simply load it with `library(...)` — for example `library(tidyverse)`.

## 5. Check that it all works

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In the **01\_Install\_R\_and\_RStudio/** folder, open `installation_test.R` in RStudio and run it (click **Source**, top-right of the editor). If everything is installed you will see:

```
My R and RStudio installation is working.
```

You can also type these four lines in the Console to check R itself:

```
2 + 2  
x <- c(10, 12, 14, 16)  
mean(x)  
library(dplyr)
```

**STUCK?** See the troubleshooting section in `Install_R_and_RStudio.md`, or email Vương Mỹ Lượng at [myluong.1710@gmail.com](mailto:myluong.1710@gmail.com). It is fine to arrive with setup half-working — there will be help on the first evening.

## 6. Open the course project

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The pre-course pack is an **RStudio Project**. Opening it points RStudio at the right folder so every file path in the lessons just works.

1. Open the **Course\_Preparation** folder on your computer.
2. Double-click `Clinical_Data_Analysis_PreCourse.Rproj`. RStudio opens with the project loaded.
3. You will see the project name at the top-right of RStudio. From now on, always open the course this way.

**WHY IT MATTERS** Because you opened the project, a path like `read_csv("Data/clinical_data_clean.csv")` finds the data automatically — you never have to edit a file path.

## 7. How to open and run the lessons

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The lessons are files you open inside RStudio. There are two kinds:

- **.R files** (scripts) — plain R code. The exercises and solutions are .R files.
- **.Rmd files** (R Markdown) — explanation and code together. The six lessons are .Rmd files.

### Open a file

Use the **Files** pane (bottom-right) to browse to a folder, then click a file to open it in the editor (top-left).

### Run one line

Click on a line of code, then press **Ctrl+Enter** (Windows) or **Cmd+Enter** (Mac). The result appears in the Console below.

### Run several lines

Select the lines with your mouse, then press **Ctrl+Enter** / **Cmd+Enter**. They run in order.

### See a whole lesson with its output (Knit)

For an .Rmd lesson, click the **Knit** button at the top of the editor. RStudio runs everything and produces a tidy document with the code, its output, and the figures. This is the easiest way to read a lesson.

**IF CODE SEEMS STUCK** A red Stop sign appears in the Console while code runs. Click it (or press Esc in the Console) to stop.

## 8. Your study path for the week

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Work through the pack in this order — about 2–3 hours in total, spread over a few short evenings.

1. Install and test (Steps 2–5 above).
2. Open the project (Step 6) and read **02\_Getting\_Started\_with\_RStudio**.
3. Do the five lessons in order: **03\_R\_Basics** → **04\_Data\_Management** → **05\_Exploratory\_Data\_Analysis** → **06\_Statistical\_Tests** → **07\_Regression**.
4. After a lesson, try the matching exercise in **Exercises/**, then check **Solutions/**.
5. Keep **Cheat\_Sheets/** and **Guides/** open for quick reference.

**HOW TO STUDY WELL** Type the code yourself instead of copying — it sticks better. Change a number and re-run to see what happens; you cannot break anything. Nothing here is graded; the goal is confidence.

## You are ready when...

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- R and RStudio are installed and `installation_test.R` prints the success message.
- You can open the .Rproj and load the dataset with `read_csv`.
- You can run one line and several lines of code.
- You have looked at R Basics and Data Management.

See you on **8 September**. — Vương Mỹ Lượng & Bernard, Neudata · #ClearDataClearImpact